

Section A: Conceptual Understanding

1. What is Ethical AI, and why is it essential in modern technological systems?

Ethical AI means creating and using AI in a way that is fair and responsible. It is important because AI can affect people's daily lives.

2. List and explain any four major ethical principles that guide AI development.

Fairness means AI should treat everyone equally. Transparency means people should know how AI works. Privacy means personal information should be protected. Accountability means developers are responsible for what their AI does.

3. What is the main objective of the IEEE Ethically Aligned Design framework?

The goal is to make sure AI is designed to help people while respecting human rights and ethical values. It encourages developers to think about the impact of AI on society.

4. Define the term algorithmic bias and provide one real-world example.

Algorithmic bias happens when an AI system gives unfair results because of biased data or programming. For example, when hiring, AI may choose more male applicants than female applicants depending on the data it receives.

5. Explain the difference between transparency and explainability in AI.

Transparency means people know how an AI system is built and used. Explainability means the AI can explain why it made a certain decision.

6. What role does accountability play in the ethical use of AI systems?

Accountability means someone is responsible if an AI system makes a mistake or causes harm. This helps people trust AI and makes companies fix problems.

7. Identify three international organizations involved in developing AI ethics guidelines.

UNESCO, IEEE, and the OECD all create guidelines for ethical AI. They help countries and companies use AI responsibly.

8. What does UNESCO's Recommendation on the Ethics of Artificial Intelligence emphasize?

It focuses on protecting human rights, fairness, and privacy. It also encourages AI to benefit everyone in society.

9. What are OECD's five key principles for trustworthy AI?

AI should benefit people, respect human rights, be transparent, be safe and secure, and have accountability.

10. How does the EU AI Act classify and regulate AI systems based on risk?

The EU AI Act places AI systems into different risk levels. High-risk AI must follow stricter rules than low-risk AI.

Section B: Applied and Scenario-Based

11. An AI recruitment tool consistently prefers male candidates. Which ethical principles are violated?

This violates fairness and equal opportunity. It also shows discrimination against female applicants.

12. A self-driving car must choose between hitting one pedestrian or five. Which ethical dilemma does this represent?

This is called the trolley problem. It asks how AI should choose between two harmful outcomes.

13. In predictive policing, how can bias be introduced into the algorithm's training data?

If the training data already contains unfair police records, the AI will learn those same biases. This can lead to unfair decisions.

14. How might a "human-in-the-loop" system reduce ethical risks in automated decision-making?

A person reviews important AI decisions before they are final. This helps catch mistakes and prevent unfair outcomes.

15. Describe how data privacy laws such as GDPR affect the design of AI systems.

These laws require AI systems to protect personal information and collect only necessary data. They also give people more control over their data.

16. A generative AI creates a fake video of a politician. What ethical issues arise here?

The fake video could spread false information and damage the person's reputation. It can also reduce public trust.

17. What are the ethical implications of AI-powered surveillance systems in public spaces?

They may invade people's privacy and make people feel constantly watched. There is also a risk of unfair targeting.

18. How can organizations ensure transparency in AI-driven financial decisions?

They should explain how the AI makes decisions and keep records of the process. Customers should be able to ask questions or appeal decisions.

19. Discuss the ethical concerns surrounding AI in healthcare diagnostics.

AI can make mistakes that affect a patient's health. It is also important to protect patient privacy and prevent bias.

20. What measures can be implemented to make AI datasets more representative and fair?

Organizations should use data from many different groups of people. They should also test datasets often to find and remove bias.

Section C: Reflective and Future-Oriented

21. Do you think AI can ever be completely unbiased? Justify your answer.

No, because AI learns from human data, which can already contain bias. However, developers can reduce bias through testing and improvements.

22. How can ethical frameworks ensure responsible innovation in AI?

They give developers rules and guidelines for creating safe and fair AI. This helps protect people while encouraging new technology.

23. Should governments enforce stricter AI regulations, or should industries self-regulate? Discuss.

I think governments should create basic laws while companies follow ethical standards. Both are needed to keep AI safe.

24. What ethical challenges might Quantum AI pose in the future?

Quantum AI could make it easier to break current security systems. It may also create powerful AI that is harder to control.

25. How might Generative AI reshape intellectual property rights?

It raises questions about who owns AI-generated content. It can also create work that closely copies other people's ideas.

26. What are the moral responsibilities of AI developers toward end-users?

Developers should create AI that is fair, safe, and respects privacy. They should also fix problems when they are discovered.

27. How can AI sustainability contribute to global ethical goals?

Using energy-efficient AI helps reduce pollution and saves resources. This supports a healthier environment for the future.

28. What future jobs might emerge to address ethical issues in AI systems?

New jobs may include AI Ethics Officer, AI Auditor, and AI Compliance Manager. These professionals will help make sure AI is used responsibly.

29. In your opinion, what will be the most pressing AI ethics issue in 2035, and why?

I think misinformation will be the biggest issue because AI will create more realistic fake content. This could make it difficult to know what information is true.

30. Reflect on your role as a future professional—how would you ensure ethical practices in AI development?

I would test AI for bias, protect people's privacy, and follow ethical guidelines. I would also make sure AI decisions are fair and easy to understand.